

Tanvir Ahmed Khan

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[Research Portfolio](#) • [Google Scholar](#) • [GitHub](#) • [ORCID](#)

Education

North South University

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

2021 – 2025

- **CGPA: 3.80/4.00** | *Summa Cum Laude*
- Relevant Coursework: Deep Learning, Pattern Recognition, Natural Language Processing, Numerical Analysis

Research Interests

Primary: machine memory, retrieval, and evidence-aware reasoning. Additional interests include multimodal learning; predictive representation learning for long-video understanding; prior and applied work in 3D vision, neural rendering, and speech-driven visual systems; efficient inference; and low-resource NLP.

Current Research

ERL — Evidence-Aware Retrieval and Memory

Active research; manuscript in preparation. Uses **paired interventions** and **matched-budget evaluation** to distinguish **routing utility** from **direct evidence utility** in multi-hop question answering. The current work is a controlled, falsification-oriented analysis after learned routing did not pass its registered gate.

2026 – Present

[Research dossier](#)

Publications

Peer-Reviewed Publications and Manuscripts

- [1] Mahir Shahriar Tamim, Md. Samiul Alim, **Tanvir Ahmed Khan**, Maisha Rahman, and Md Musfique Anwar. “Ensemble and Temporal Feature-Based Framework for Rainfall Classification in Bangladesh.” *PLOS ONE*, 21(3): e0342646, 2026. [DOI](#).
- [2] Mahir Shahriar Tamim, Sharjil Khan, Md. Samiul Alim, **Tanvir Ahmed Khan**, Shafin Rahman, and Nabeel Mohammed. “CAT-GS: Balanced Multimodal Learning via Calibrated Gating and Fusion Surgery.” Manuscript under minor revision at *Neurocomputing*, 2026. [arXiv](#).
- [3] Md. Samiul Alim, Mahir Shahriar Tamim, **Tanvir Ahmed Khan**, Sharjil Khan, Rafia Ferdous Duti, Shahriyar Zaman Ridoy, and Mohammad Ali Moni. “Can LLMs Follow the Pulse of a Crisis? Evaluating Crisis Sentiment in Bangladesh’s July Uprising.” *AACL-IJCNLP 2026*, Main Conference, accepted, 2026. [arXiv](#).
- [4] Md. Samiul Alim, Mahir Shahriar Tamim, Maisha Rahman, **Tanvir Ahmed Khan**, and Md Mushfique Anwar. “When a Nation Speaks: Machine Learning and NLP in People’s Sentiment Analysis During Bangladesh’s 2024 Mass Uprising.” In *2025 28th International Conference on Computer and Information Technology (ICCIT)*, pp. 413–418, IEEE, 2025. [DOI](#).

Research Experience

Machine Intelligence Lab (MILab) | North South University

Mar 2025 – Present

- **Research Assistant** under [Dr. Nabeel Mohammed](#) and [Dr. Shafin Rahman](#), working on **multimodal systems**, low-resource NLP, and speech-driven visual systems.
- Built and evaluated streaming conversational-avatar pipelines, including **representation bridges**, synchronized multi-GPU execution, facial-motion recovery, and **renderer distillation**.
- Performed **TensorRT** validation and **Jetson AGX Orin** profiling; renderer acceleration shifted the measured bottleneck to upstream speech generation. Upstream models were external, portions are closed source, and end-to-end real-time operation was not established.

Selected Research and Engineering Projects

PNEUMA — Video Memory for Long-Form Question Answering

Completed thesis / discontinued

Undergraduate thesis integrating SQL, vector, and temporal-graph retrieval for long-video memory. Discontinued after an audit exposed unresolved grounding and correctness limits.

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AirFLOW — Auditable Aerodynamic Analysis Workbench

Recent / early-stage

Early-stage collaboration with an aerospace practitioner and MIST graduate: STL and lifting-surface audit, standard horseshoe VLM, and an OpenFOAM case handoff without CFD physics validation.

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DynApex: Dynamic CPU/GPU Parallelization

Prototype / 2024

Engineering prototype for dependency-aware workload partitioning and OpenMP-based CPU/GPU target execution.

[GitHub](#)

Industry Experience

Tuctuc | Spain (Remote)

Jan 2022 – Jun 2026

- **Junior Software Engineer.** Maintained and extended Odoo-based ERP systems and developed cross-platform mobile applications using React Native.

Technical Skills

Machine Learning & Deep Learning: PyTorch, Hugging Face Transformers, Knowledge Distillation, PEFT/LoRA

3D Vision & Neural Rendering: 3D Gaussian Splatting, FLAME, 3D Morphable Models, facial motion modeling

Multimodal, Retrieval & NLP: vision-language models, predictive video learning, retrieval-augmented generation, FAISS, Knowledge Graphs, SQL

Systems & Deployment: multi-GPU inference, CUDA, Docker, Linux, TensorRT, NVIDIA Jetson AGX Orin

Programming: Python, C++, JavaScript, React Native

Languages, Honors & Activities

Bangla — Native | English — C1 (IELTS Academic 7.0)

July 2026

Academic Scholarship, North South University (2023) • **Participant**, ICPC Asia Dhaka Preliminary Contest (2021)